CHAPTER 2:

Wave Propagation – Guided Waves

Contents

Introduction

You have already studied propagation of plane waves in free space and their interaction with dielectric and metallic interfaces. We are now concerned here with the propagation of guided electromagnetic waves. That is, waves which are forced by conductors and/or dielectrics to carry their energy in a particular direction and mainly through a limited region in the transverse plane. In general, any configuration of conductors and/or dielectrics that supports a wave propagating alongside and with limited transverse extent is called a waveguide. A simple example of this situation is the well-known case of the two-wires transmission line, where the electromagnetic wave propagates along the lines and the fields have significant values only in a limited region around the wires, although they extend to infinity. Other examples are hollow metal pipes, dielectric rods (optical fibres), and so on.

In the following sections it will be shown that electromagnetic fields can exist in a waveguide in a variety of forms called modes of propagation. This will be demonstrated firstly by using superpositions of plane waves and showing that these can exist in the waveguide and secondly, by finding the solutions to the wave equation for the cases of rectangular and circular waveguides.

Modes of Propagation

Transverse Electromagnetic, (TEM) waves

In the previous section you studied plane waves (part 1 of the course) and in those the electric and magnetic fields were both totally contained in the plane transverse to the direction of propagation of the wave, that is $E_z = H_z = 0$ where z is the propagation direction. The same is observed in transmission lines like the common two-wires line and the coaxial cable. Both electric and magnetic fields are purely transverse. For this reason, this type of wave is called a *TEM or transverse electromagnetic wave*. Fields and waves with this property are certainly one of the solutions to the wave equation for the transmission line but by no means they are the only one. That is, there are other forms than this, which the electromagnetic fields can assume in waves propagating along transmission lines. The TEM wave is certainly the commonest mode of propagation in these structures from the practical point of view of utilization of the line.

In this case it can be easily shown that the electric field (and also the magnetic field can be derived from a scalar potential which satisfies the 2-dimensional Laplace equation:

$$\nabla_t^2 \Phi = 0\,, \qquad E_t = -\nabla_t \Phi$$

Example 2.1

A coaxial line

Let's consider a coaxial line with conductors of radii a and b and a voltage difference V_0 between conductors.

In polar coordinates, Laplace's equation is:

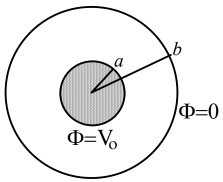

Fig. 2.1 Coaxial line cross-section

$$\nabla_t^2 \Phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \Phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \Phi}{\partial \varphi^2} = 0$$

Because of the symmetry of the structure, we look for a function Φ independent of the angle φ:

$\Phi = \Phi(r)$, then the previous equation reduces to:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \Phi}{\partial r}\right) = 0 \Rightarrow r\frac{\partial \Phi}{\partial r} = C_1\ (Const.) \Rightarrow \Phi = C_1 \ln r + C_2$$

With the boundary conditions:

$$\left.\begin{matrix}\Phi = V_0 & \text{at } r = a \\ \Phi = 0 & \text{at } r = b\end{matrix}\right\} \text{ gives } \left.\begin{matrix}V_0 = C_1 \ln a + C_2 \\ 0 = C_1 \ln b + C_2\end{matrix}\right\} \text{ then } \left\{\begin{matrix}C_2 = -C_1 \ln b \\ C_1 = V_0/\ln(a/b)\end{matrix}\right.$$

So, finally,
$$\Phi = V_0 \frac{\ln(b/r)}{\ln(b/a)}$$

The electric field is then: $$\vec{\mathbf{E}} = -\nabla_t \Phi = -\left(\frac{\partial \Phi}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial \Phi}{\partial \varphi}\hat{\boldsymbol{\varphi}}\right) = -\frac{\partial \Phi}{\partial r}\hat{\mathbf{r}} = \frac{V_0}{\ln(b/a)}\frac{\hat{\mathbf{r}}}{r}$$

and the magnetic field: $$\vec{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \vec{\mathbf{E}} = \frac{V_0}{\eta \ln(b/a)}\frac{\hat{\boldsymbol{\varphi}}}{r}$$

That completes the description of the fields in this structure. As expected, both $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$ are purely transverse: – they constitute a TEM wave.

The power flow in the TEM mode is given by:

$$P = \frac{1}{2}\mathrm{Re}\left\{\iint \vec{\mathbf{E}} \times \vec{\mathbf{H}}^* \cdot \hat{\mathbf{z}}\, r\,dr\,d\varphi\right\} = \frac{V_0^2 \pi}{\eta \ln(b/a)} \quad [\mathrm{W}]$$

The propagation constant is $\gamma = jk_0$ if the dielectric is air and then, the phase velocity is simply c.

For the common transmission line formed by two parallel conductors a similar analysis (but rather more complicated mathematically) can be carried out to find the fields around the line. Also in this case the commonest form of these fields will constitute a TEM wave, with both $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$ purely transverse.

In all TEM waves all fields can be completely calculated once the potential function Φ has been found, we can then describe the behaviour of the transmission line by the variation of this potential along the axial direction, and in particular by its value at the conductors which constitute the line, the familiar $V(z)$ wave. In the same form, if we calculate the line integral along any closed path surrounding one of the conductors we will have (by Ampère's law) the total current flowing through that conductor, the current wave $I(z)$.

We will study next the case of field confinement (waveguiding) produced by two infinite parallel plates, which can be considered as an extension of the well-known two-conductor transmission line. This case will be analysed first considering the fields as if they were produced by a superposition of plane waves.

Waveguide Modes: Plane Waves Approach

Parallel plates: – TEM waves.

Let's first consider a plane wave travelling in the z direction and with the electric field polarized along the y axis:

$$\vec{\mathbf{E}} = E_0 e^{j(\omega t - k_0 z)} \hat{\mathbf{y}} \qquad (2.1a)$$

$$\vec{\mathbf{H}} = \frac{E_0}{\eta_0} e^{j(\omega t - k_0 z)} \hat{\mathbf{z}} \times \hat{\mathbf{y}} \qquad (2.1b)$$

$$= -\frac{E_0}{\eta_0} e^{j(\omega t - k_0 z)} \hat{\mathbf{x}}$$

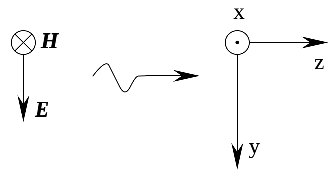

Fig. 2.2 Field polarization

Both fields are uniform and perpendicular to the direction of propagation.

Suppose now that we put two infinite, perfectly conducting plates at the planes $y = \pm\ b/2$. This should not alter the fields or modify the wave in any way since on the surface of the conductors: $\vec{\mathbf{E}}_{tan} = 0$, $\vec{\mathbf{H}}_{normal} = 0$.

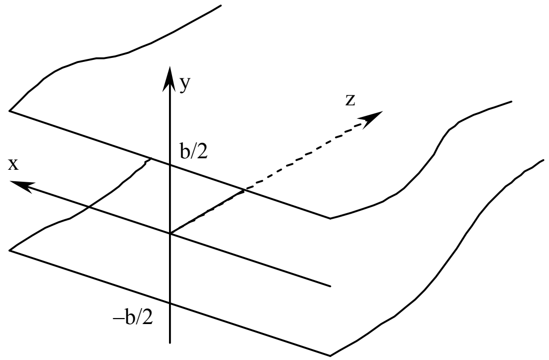

Fig. 2.3 Two parallel conducting plates

But by placing these conducting plates there, the three regions of space will be totally isolated and independent, and then we can concentrate in studying what happens in the space between them. Furthermore, we may assume that the fields outside are zero.

This shows that the fields given by (2.1) can exist as a propagating wave in the space between the plates, and then we have a confined wave.

From (2.1) we see that the electric field is perpendicular to the plates, the magnetic field is parallel to them and both are perpendicular to the direction of propagation. This is a TEM wave since there is no longitudinal component of $\vec{\mathbf{E}}$ or $\vec{\mathbf{H}}$.

The propagation constant of this wave is $\gamma = jk_0$, the same as in the open space and then, so it is its phase velocity: $v_p = \omega/k_0 = c$.

The value of k_0 is dependent of the frequency: $k_0 = \omega\sqrt{\mu_0\varepsilon_0} = 2\pi f\sqrt{\mu_0\varepsilon_0}$ and is not modified by the presence of the conducting plates.

But, as said before, this is not the only form the electromagnetic fields can assume in this type of guide. We will see next another forms.

Parallel plates: – TE and TM modes

Let's consider now the situation produced by the interference of two plane waves propagating at an angle with the electric field polarized in the direction of y as shown in Fig. 3.4.

The electric and magnetic fields of both waves are perpendicular to their directions of propagation but as they travel at an angle respect to the z axis, there will be components of $\vec{\mathbf{H}}$ in the direction of x and z. The $\vec{\mathbf{E}}$ field of both waves is always parallel to the y-axis.

We will then have, omitting the factor $\exp(j\omega t)$:

$$\vec{\mathbf{E}}_1 = E_0 e^{-jk_0\hat{n}_1\cdot\vec{r}}\,\hat{\mathbf{y}}$$

and $\hat{n}_1\cdot\vec{r} = (-\sin\theta\,\hat{x} + \cos\theta\,\hat{z})\cdot(x\hat{x} + z\hat{z})$

$$= -x\sin\theta + z\cos\theta$$

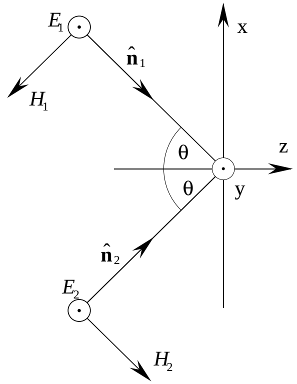

Fig. 2.4 Two incident waves

which gives for the electric field:

$$\vec{\mathbf{E}}_1 = E_0 e^{jk_0 x\sin\theta} e^{-jk_0 z\cos\theta}\,\hat{\mathbf{y}} \qquad (2.2a)$$

and for the magnetic field:

$$\vec{\mathbf{H}}_1 = \frac{1}{\eta_0}\hat{n}_1 \times \vec{\mathbf{E}}_1 = \frac{E_0}{\eta_0} e^{jk_0 x\sin\theta} e^{-jk_0 z\cos\theta}(-\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{z}}) \times \hat{\mathbf{y}} \qquad (2.2b)$$

For simplicity, we define:

$$k_x = k_0\sin\theta, \quad k_z = k_0\cos\theta \quad (\Rightarrow\ k_0^{\,2} = k_x^{\,2} + k_z^{\,2}) \qquad (2.3)$$

Now, separating the two components of the magnetic field in (2.2b) gives:

$$H_{1x} = -\frac{E_0}{\eta_0}\cos\theta e^{jk_x x} e^{-jk_z z} \qquad (2.4a)$$

and

$$H_{1z} = -\frac{E_0}{\eta_0}\sin\theta e^{jk_x x} e^{-jk_z z} \qquad (2.4b)$$

Similarly for the second wave we have: $\vec{\mathbf{E}}_2 = E_0 e^{-jk_0\hat{n}_2\cdot\vec{r}}\,\hat{\mathbf{y}}$

but now $\hat{n}_2\cdot\vec{r} = (\sin\theta\,\hat{x} + \cos\theta\,\hat{z})\cdot(x\hat{x} + z\hat{z}) = x\sin\theta + z\cos\theta$

then:

$$\vec{\mathbf{E}}_2 = E_0 e^{-jk_x x} e^{-jk_z z}\,\hat{\mathbf{y}} \qquad (2.5a)$$

$$H_{2x} = -\frac{E_0}{\eta_0}\cos\theta e^{-jk_x x} e^{-jk_z z} \tag{2.5b}$$

and
$$H_{2z} = \frac{E_0}{\eta_0}\sin\theta e^{-jk_x x} e^{-jk_z z} \tag{2.5c}$$

Finally, the total electromagnetic field produced by the superposition (interference) of the two plane waves has the three components E_y, H_x and H_z.

$$E_y = E_1 + E_2 = 2E_0 \cos k_x x e^{-jk_z z} \tag{2.6a}$$

$$H_x = H_{1x} + H_{2x} = -2\frac{E_0}{\eta_0}\cos\theta \cos k_x x e^{-jk_z z} \tag{2.6b}$$

$$H_z = H_{1z} + H_{2z} = -2j\frac{E_0}{\eta_0}\sin\theta \sin k_x x e^{-jk_z z} \tag{2.6c}$$

We can note immediately that the resultant fields (equations (2.6)) propagate like a wave in the direction of the z axis and with a wavenumber (or phase constant) equal to $k_z = k_0\cos\theta$. The amplitude of this wave is independent of y, but depends on x and we can observe that there are positions of zero amplitude: There is an <u>interference pattern</u>. In particular, if $k_x x$ $(= k_0 x\sin\theta) = \pi/2 \pm n\pi$ then E_y and H_x are zero for all y, z and t. And the same will happen at any x for which $\cos k_x x = 0$ (that is, for $x = (2n+1)\pi/2k_x$, for integer n positive, 0 or negative.
The contour lines in Fig. 2.5 show the values of the total (real) electric field as in (2.6a). The interference effect is clearly visible. The solid lines correspond to positions where the electric field is zero.

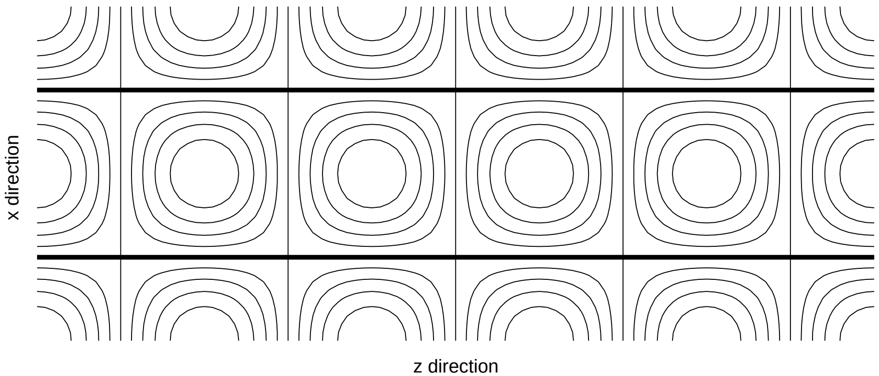

Fig. 2.5 Interference pattern produced by two plane waves

Suppose now that we put perfectly conducting sheets at any two such places, like $x = \pm\,\pi/2k_x$. The conductors need not modify the field distribution or modify the waves in any way since the boundary conditions are satisfied in those positions:

$$E_{tan} = E_y = 0, \qquad B_{normal} = \mu_0 H_x = 0$$

at the conductor surfaces.

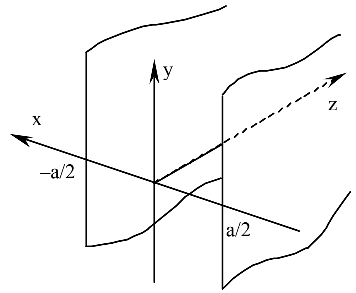

Fig. 2.6 Parallel plates at $a = \pi/k_x$.

The previous analysis shows that apart from the TEM wave studied earlier, the fields given by (2.6) can also exist as a propagating wave between the plates.
Putting back the term $e^{j\omega t}$, we can rewrite (2.6) as:

$$E_y = E_0 \cos k_x x e^{j(\omega t - k_z z)} \tag{2.7a}$$

$$H_x = H_{x0}\cos k_x x e^{j(\omega t - k_z z)} \qquad \left(\text{with } H_{x0} = -2\frac{E_0}{\eta_0}\cos\theta\right) \tag{2.7b}$$

$$H_z = jH_{z0}\sin k_x x e^{j(\omega t - k_z z)} \qquad \left(\text{with } H_{z0} = -2\frac{E_0}{\eta_0}\sin\theta\right) \tag{2.7c}$$

A wave like this is called a ***TE wave***, *transverse electric wave* or an *H-wave* since only the electric field is transverse (it lies on a plane transverse to z).
Similarly, if we consider plane waves with the electric field in the plane x-z instead of that in (2.1), the resultant fields would have components E_x, E_z and H_y, a so called ***TM*** (*transverse magnetic*) wave or ***E-wave***.

Having succeeded in confining the electromagnetic fields to the region: $|x| \le \frac{\pi}{2k_x} = \frac{a}{2}$,
we can now equally regard the fields as due to waves bouncing off the two conducting plates. From this viewpoint, the correct spacing between conductors corresponds to the condition that two successive reflections give a wave exactly in phase with the original and we can note that this means that at a given frequency, the wave can only exist if the correct relationship between k_x and a is observed: $k_x = (\pi + 2n\pi)/a$.

The phase constant is $k_z = \sqrt{k_0^2 - k_x^2}$ (see (2.7)) and we can observe that if at a given frequency f_c, k_0 (given by $2\pi f_c\sqrt{\mu_0\varepsilon_0}$) is equal to k_x no propagation is possible since k_z will be exactly zero. Because of this, this frequency f_c is called the ***<u>cutoff frequency</u>***. Even more, at frequencies lower than f_c, k_z becomes imaginary and the propagation factor becomes exp(–$|k_z|z$), (with a purely real exponent) and the amplitude decays exponentially along the z-direction without oscillation. Then, propagation can only exist at frequencies greater than f_c.

Modes of Propagation

Now that we have fixed the plates at positions $\pm\, a/2$, let's imagine that we decide to change the angle of incidence θ of the waves. You will notice that the positions along the x axis where the boundary conditions are now satisfied do not, in general, coincide with the positions of the plates. However, for discrete values of θ this condition will be true: this will occur for values of θ such that $\sin\theta = (\pi/2 \pm n\pi)/k_0 a$. There are infinite such values of θ. For each of them, a different field distribution will exist between the plates. These are called ***'modes of propagation'***. One of them is that already described in equations (2.7).

Rectangular Waveguide

We can see that two more conducting plates can be placed at positions $y = \pm b/2$ (with arbitrary b) since (as with the wave in the TEM case) the boundary conditions on these new plates are already satisfied by the fields of (2.7):

$$E_{tan} = (E_x \text{ or } E_z) = 0; \qquad H_{normal} = (H_y) = 0$$

Therefore, we have found expressions for one of the forms of the electromagnetic fields which constitute a wave totally confined to the region inside the rectangular hollow pipe of cross-section dimensions a and b, or ***rectangular waveguide***. These are given by equations (2.7). A more general and complete picture is obtained by finding all possible solutions of the wave equation with the boundary conditions imposed by the waveguide walls.

Waveguide Modes: Solution of the Wave Equation

Fundamentals

A full description of wave propagation inside a hollow pipe and in general, around any uniform waveguiding structure (a two-wire transmission line for example) is based in the direct solution of the wave equation (or Helmholtz equation) for phasor fields:

$$\nabla^2 \vec{\mathbf{E}} = -k^2 \vec{\mathbf{E}} \qquad \text{or} \qquad \nabla^2 \vec{\mathbf{H}} = -k^2 \vec{\mathbf{H}} \tag{2.8}$$

where $k^2 = \omega^2 \mu\varepsilon$, with the boundary conditions imposed by the conducting walls in the waveguide. This, of course, is valid for any arbitrary cross-section (and not necessarily only for the rectangular guide).

We can formally separate the three derivatives in the term $\nabla^2 \vec{\mathbf{E}}$ writing:

$$\nabla^2 \vec{\mathbf{E}} = \nabla_t^2 \vec{\mathbf{E}} + \frac{\partial^2 \vec{\mathbf{E}}}{\partial z^2} \tag{2.9}$$

where the two transverse derivatives are grouped together defining: $\nabla_t^2 \vec{\mathbf{E}} = \dfrac{\partial^2 \vec{\mathbf{E}}}{\partial x^2} + \dfrac{\partial^2 \vec{\mathbf{E}}}{\partial y^2}$

If we assume that the wave propagates in the direction of the z-axis, the fields will have a z-dependence of the form: $\exp(-\gamma z)$ and the second derivative respect to z is simply $\gamma^2 \vec{\mathbf{E}}$.

Therefore, Helmholtz equations can be written in the form:

$$\boxed{\begin{aligned} \nabla_t^2 \vec{\mathbf{E}} &= -\left(\gamma^2 + k^2\right)\vec{\mathbf{E}} \\ \nabla_t^2 \vec{\mathbf{H}} &= -\left(\gamma^2 + k^2\right)\vec{\mathbf{H}} \end{aligned}} \tag{2.10}$$

Also, if we re-write Maxwell's curl equations, explicitly separating their three components, we have three scalar equations from each of them:

$\nabla \times \vec{\mathbf{E}} = -j\omega\mu \vec{\mathbf{H}}$	$\nabla \times \vec{\mathbf{H}} = j\omega\varepsilon \vec{\mathbf{E}}$
$a)\ \dfrac{\partial E_z}{\partial y} + \gamma E_y = -j\omega\mu H_x$	$d)\ \dfrac{\partial H_z}{\partial y} + \gamma H_y = j\omega\varepsilon E_x$
$b)\ -\gamma E_x - \dfrac{\partial E_z}{\partial x} = -j\omega\mu H_y$	$e)\ -\gamma H_x - \dfrac{\partial H_z}{\partial x} = j\omega\varepsilon E_y$
$c)\ \dfrac{\partial E_y}{\partial x} - \dfrac{\partial E_x}{\partial y} = -j\omega\mu H_z$	$f)\ \dfrac{\partial H_y}{\partial x} - \dfrac{\partial H_x}{\partial y} = j\omega\varepsilon E_z$

(2.11)

We can use this form of the equations to write all transverse components of $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$: E_x, E_y, H_x and H_y in terms of the longitudinal components. That is, we do not need all components to describe completely the fields. Only the two longitudinal components will be necessary and the rest can be obtained from them:

$$\text{(a)}\quad E_x = -\frac{j}{k_c^2}\left(\beta\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right) \qquad \text{(c)}\quad H_x = \frac{j}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial y} - \beta\frac{\partial H_z}{\partial x}\right)$$

$$\text{(b)}\quad E_y = \frac{j}{k_c^2}\left(-\beta\frac{\partial E_z}{\partial y} + \omega\mu\frac{\partial H_z}{\partial x}\right) \qquad \text{(d)}\quad H_y = -\frac{j}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial x} + \beta\frac{\partial H_z}{\partial y}\right) \tag{2.12}$$

where we have written $\gamma = j\beta$ and defined: $k_c^2 = k^2 - \beta^2$.

Exercise: Derive expressions (2.11) and (2.12).

This is an important fact that means that for the parallel plates waveguide and for any hollow-pipe waveguide with any shape of cross-section and in general for any uniform guide, (including multi-conductor transmission lines) we do not need to solve the vectorial equations (2.10) but instead we only need to find the scalar quantities E_z and H_z.

From equation (2.10) we can see that the longitudinal components satisfy the equations:

$$\nabla_t^2 E_z = -k_c^2 E_z \tag{2.13a}$$

$$\nabla_t^2 H_z = -k_c^2 H_z \tag{2.13b}$$

Then, for an arbitrary uniform waveguiding structure along the z-axis, we need only solve the scalar equations (2.13) for the longitudinal components and the transverse components can all be obtained from equations (2.12).

Furthermore, if the fields are either of the TE or TM type, only one of the scalar functions E_z or H_z will be needed to describe completely all the fields.

Hollow pipe Waveguides

This type of waveguide, consisting of a hollow conducting cylinder, cannot support a TEM wave, since these waves have transverse variations like static fields and static fields cannot exist inside a region bounded by a single conductor. A more rigorous explanation follows:

Let's assume that a TEM wave exists inside this waveguide. From Maxwell's curl equations (2.11) we have:

$$\nabla_t \times \vec{\mathbf{E}}_t = \frac{\partial \vec{\mathbf{E}}_y}{\partial x} - \frac{\partial \vec{\mathbf{E}}_x}{\partial y} = -j\omega\mu H_z \hat{\mathbf{z}}$$

but in a TEM wave $H_z = 0$ and then, $\nabla_t \times \vec{\mathbf{E}}_t = 0$. In consequence, $\vec{\mathbf{E}}_t$ can be written as the gradient of a scalar function: $\vec{\mathbf{E}}_t = -\nabla_t \phi(x, y)$.

Now, $\nabla \cdot \vec{\mathbf{E}} = 0$ (Gauss law), but as $E_z = 0$ since the wave is TEM, we have that $\nabla_t \cdot \vec{\mathbf{E}}_t = 0$ (containing the transverse derivatives of the transverse components) and we get that ϕ should satisfy $\nabla_t^2 \phi = 0$, equation that characterizes static fields in two dimensions. If we now impose the boundary condition ϕ = Const. on the whole boundary, we find that the field must be zero everywhere.

Another way to see this is as follows: A TEM mode has both $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$ lying in the transverse plane only. Consequently, the lines of $\vec{\mathbf{H}}$ lie entirely on the transverse plane. Also, as $\nabla \cdot \vec{\mathbf{H}} = 0$, the lines of $\vec{\mathbf{H}}$ are closed loops in the transverse plane (perpendicular to the axis of the guide). Then, according to Ampère's law, the magnetomotive force $\oint \vec{\mathbf{H}} \cdot d\vec{l}$ must be equal to a current across the loop (an axial current). For this current to exist we need either a conductor in the axial direction to support conduction currents $\vec{\mathbf{J}}$ or in case of displacement currents, an electric field in the axial direction. In consequence, TEM waves are not possible in hollow pipes.

Consequently, the modes of propagation that exist inside hollow pipes should have at least one of the two components E_z, H_z present.

We will now study TE and TM waves in hollow pipes of any cross-section.

Transverse electric TE waves

TE waves have $E_z = 0$ and satisfy equation (2.13b) for H_z:

$$\nabla_t^2 H_z = \frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} = -k_c^2 H_z \tag{2.14}$$

Now, from equations (2.12c) and (2.12d) with $E_z = 0$ we can write:

$$\vec{\mathbf{H}}_t = -\frac{j\beta}{k_c^2} \nabla_t H_z \tag{2.15}$$

also, from (2.11a) and (2.11b) with $\gamma = j\beta$:

$$\vec{\mathbf{E}}_t = -\frac{\omega\mu_0}{\beta}\left(\hat{\mathbf{z}} \times \vec{\mathbf{H}}_t\right) \tag{2.16}$$

so both transverse components can be derived from H_z. and we only need to solve the equation (2.14) for H_z. **<u>Exercise</u>**: Prove (2.15) and (2.16)

We can observe that the ratio between magnitudes of transverse magnetic and electric components is given by:

$$\frac{\omega\mu_0}{\beta} = \frac{k_0}{\beta}\eta_0 = Z_h \tag{2.17}$$

Z_h has dimensions of impedance and is called wave impedance of TE modes.

Summary for TE or H waves:

First find a solution for H_z using equation (2.14)
– then find $\vec{\mathbf{H}}_t$ using (2.15), and $\vec{\mathbf{E}}_t$ using (2.16).

The individual components of $\vec{\mathbf{H}}_t$ and $\vec{\mathbf{E}}_t$ can also be found directly using (2.12).

Transverse magnetic (TM) waves

In this case $H_z = 0$ and E_z satisfies the equation:

$$\nabla_t^2 E_z + k_c^2 E_z = 0 \tag{2.18}$$

Again, using (2.11) and (2.12) we can write:

$$\vec{\mathbf{E}}_t = -\frac{j\beta}{k_c^2} \nabla_t E_z \tag{2.19}$$

$$\vec{\mathbf{H}}_t = \frac{\omega\varepsilon}{\beta}\left(\hat{\mathbf{z}} \times \vec{\mathbf{E}}_t\right) = \frac{k_0}{\beta\eta_0}\left(\hat{\mathbf{z}} \times \vec{\mathbf{E}}_t\right) \tag{2.20}$$

and, as for TE modes, all transverse components can be obtained from the longitudinal, in this case, E_z.

The wave impedance in this case (ratio between transverse field components) is:

$$Z_e = \frac{\beta}{k_0}\eta_0 \tag{2.21}$$

Note the difference with respect to TE modes.

Summary for TM or E waves:

First find a solution for E_z using equation (2.18)
– then find $\vec{\mathbf{E}}_t$ using (2.19), and $\vec{\mathbf{H}}_t$ using (2.20).

And again, all individual components of transverse fields can be found directly using (2.12).

Rectangular waveguide

The most important of the hollow pipe waveguides is that of rectangular cross section.

We assume the guide has dimensions *a* and *b*, with perfectly conducting walls and extends indefinitely in the *z* direction.

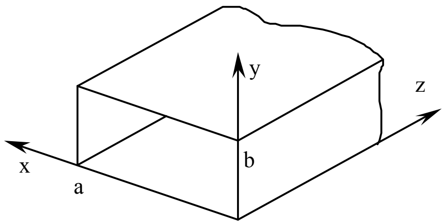

Fig. 2.7 Rectangular waveguide cross-section

TE or H waves:

For this type of waves $E_z = 0$ and H_z satisfies equation (2.14):

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + k_c^2 H_z = 0$$

which can be solved using separation of variables as in the case of the plane wave in open space.

Putting $H_z(x,y) = f(x)\, g(y)$, we have:

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + k_c^2 H_z = 0$$

which can be separated as: $\frac{d^2 f}{dx^2} + k_x^2 f = 0$ and $\frac{d^2 g}{dy^2} + k_y^2 g = 0$

where $k_x^2 + k_y^2 = k_c^2$

The solution is then:

$$H_z = fg = \left(A_1 \cos k_x x + A_2 \sin k_x x\right)\left(B_1 \cos k_y y + B_2 \sin k_y y\right) \qquad (2.24)$$

One of the boundary conditions to be satisfied on the surface of the perfectly conducting walls is $\hat{\mathbf{n}} \cdot \vec{\mathbf{H}}_t = 0$. (The **H** field is tangential to the walls)

The application of this condition is sufficient and the resultant field will also satisfy $\hat{\mathbf{n}} \times \vec{\mathbf{E}} = 0$. **Exercise**: Show that this is true.

Using (2.15), the application of this boundary condition leads to:

$$\hat{\mathbf{n}} \cdot \left(\frac{\partial H_z}{\partial x} \hat{\mathbf{x}} + \frac{\partial H_z}{\partial y} \hat{\mathbf{y}} \right) = 0 \qquad \text{on the walls.}$$

that is: $\frac{\partial H_z}{\partial x} = 0$ at $x = 0$, and $x = a$

and $\frac{\partial H_z}{\partial y} = 0$ at $y = 0$, and $y = b$

which when applied to (2.24) lead to:

$$\left(-k_x A_1 \sin k_x x + k_x A_2 \cos k_x x\right) g(y) = 0 \qquad \text{at } x = 0,\ a$$

$$f(x)\left(-k_y B_1 \sin k_y y + k_y B_2 \cos k_y y\right) = 0 \qquad \text{at } y = 0,\ b$$

and consequently, $A_2 = 0, \quad B_2 = 0$

$$k_x = \frac{n\pi}{a}, \ n = 0, 1, 2, \ldots \qquad k_y = \frac{m\pi}{b}, \ m = 0, 1, 2, \ldots \qquad (2.25)$$

Then, the general solution for TE modes in a rectangular waveguide is:

$$\boxed{\begin{array}{c} H_z = A_{nm} \cos\frac{n\pi x}{a} \cos\frac{m\pi y}{b} \\ \text{for } n, m = 0, 1, 2, \ldots \text{ but not } n = m = 0 \end{array}} \qquad (2.26)$$

and the remaining components are obtained from equations (2.12):

We can see that in this case, for each pair of values of *n* and *m* (not both simultaneously equal to 0) there is a solution for the fields. These solutions are called ***modes of propagation***, and equation (2.26) defines a general TE_{nm} mode.

The propagation constant $\gamma_{nm} = j\beta_{nm}$ (if there are no losses) for the TE_{nm} mode is given by:

$$\gamma_{nm} = j\beta_{nm} = j\sqrt{k_0^2 - k_{c,nm}^2} \qquad (2.27)$$

with the constant $k_{c,nm}$ (or cutoff wavenumber):

$$k_{c,nm}^2 = k_x^2 + k_y^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2 \qquad (2.28)$$

then, for the propagation constant we get:

$$\gamma_{nm} = j\sqrt{\left(\frac{2\pi}{\lambda_0}\right)^2 - \left(\frac{n\pi}{a}\right)^2 - \left(\frac{m\pi}{b}\right)^2} \qquad (2.29)$$

When $k_0 < k_{c,nm}$, the propagation constant γ_{nm} is real and the wave no longer propagates, the propagation factor $\exp(-\gamma z)$ shows that the fields simply decay with the distance z.

From (2.28) we can see that the frequency for which this starts to occur for the nm mode is the ***cutoff frequency*** given by:

$$f_{c,nm} = \frac{c}{2\pi} k_{c,nm} = \frac{c}{2}\sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} \tag{2.30}$$

The wave impedance is (as in (2.17)):

$$Z_{h,nm} = \frac{k_0}{\beta_{nm}}\eta_0$$

For the phase velocity, $v_p = \frac{\omega}{\beta}$ and for the propagating regime, that is for frequencies above the cutoff frequency, we have:

$$v_p = \frac{\omega}{\sqrt{k_0^2 - k_c^2}} = c\frac{1}{\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2}} = c\frac{1}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

assuming that the dielectric in the guide is air. If the waveguide is filled with a dielectric, k_0 and c in the above expressions change to k and v, the values of the wavenumber and light velocity in the dielectric.

For the group velocity, $v_g = \frac{\partial\omega}{\partial\beta}$ the following expression can be found:

$$v_g = c\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2} = c\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

Again, this expression is for frequencies above cutoff and assuming an air-filled waveguide.

In Summary: TE modes in a rectangular waveguide are characterized by:

$$E_z = 0$$

$$H_z = A_{nm}\cos\frac{n\pi x}{a}\cos\frac{m\pi y}{b} \qquad (2.26):$$

the other components of the electric and magnetic fields are obtained from (2.12), or from (2.15) and (2.16). That is:

$$H_z = A_{nm}\cos\frac{n\pi x}{a}\cos\frac{m\pi y}{b}e^{-j\beta_{nm}z} \tag{2.31}$$

$$H_x = j\frac{\beta_{nm}n\pi}{k_{c,nm}^2 a}A_{nm}\sin\frac{n\pi x}{a}\cos\frac{m\pi y}{b}e^{-j\beta_{nm}z} \qquad E_x = Z_{h,nm}H_y \tag{2.32}$$

$$H_y = j\frac{\beta_{nm}m\pi}{k_{c,nm}^2 b}A_{nm}\cos\frac{n\pi x}{a}\sin\frac{m\pi y}{b}e^{-j\beta_{nm}z} \qquad E_y = -Z_{h,nm}H_x \tag{2.33}$$

Each mode will propagate at frequencies above the cutoff frequency:

$$f_{c,nm} = \frac{c}{2\pi}k_{c,nm} = \frac{c}{2}\sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2}$$

and with a propagation constant and wave impedance given by:

$$\gamma_{nm} = j\beta_{nm} = j\sqrt{\left(\frac{2\pi}{\lambda_0}\right)^2 - \left(\frac{n\pi}{a}\right)^2 - \left(\frac{m\pi}{b}\right)^2} \qquad Z_{h,nm} = \frac{k_0}{\beta_{nm}}\eta_0$$

TM or E Waves

In this case $H_z = 0$ and all remaining field components can be derived from E_z which satisfies the equation (2.13) subject to the boundary conditions $E_z = 0$ on the waveguide walls: $x = 0, a$, $y = 0, b$.

The solution for E_z is therefore:

$$E_z = B_{nm}\sin\frac{n\pi x}{a}\sin\frac{m\pi y}{b} \quad \text{for } n, m = 1, 2, \ldots \tag{2.34}$$

Again, all remaining components are obtained from E_z using relations (2.12) or from (2.19) and (2.20).

The cutoff wavenumber is given by the same expression as for TE modes and consequently, f_c is again obtained from (2.30) and the propagation constant, γ_{nm}, from (2.29).

The expressions for phase and group velocities obtained for TE modes are also valid in this case.

Summary for TM modes:

$$H_z = 0$$

$$E_z = B_{nm}\sin\frac{n\pi x}{a}\sin\frac{m\pi y}{b} \qquad (2.34):$$

the other components of the electric and magnetic fields are obtained from (2.12), or from (2.15) and (2.16). That is:

$$E_z = B_{nm}\sin\frac{n\pi x}{a}\sin\frac{m\pi y}{b}e^{-j\beta_{nm}z} \tag{2.35}$$

$$E_x = -j\frac{\beta_{nm}n\pi}{k_{c,nm}^2 a}B_{nm}\cos\frac{n\pi x}{a}\sin\frac{m\pi y}{b}e^{-j\beta_{nm}z} \qquad H_y = -\frac{1}{Z_{e,nm}}E_x \tag{2.36}$$

$$E_y = -j\frac{\beta_{nm} m\pi}{k_{c,nm}^2 b} B_{nm} \sin\frac{n\pi x}{a}\cos\frac{m\pi y}{b} e^{-j\beta_{nm} z} \qquad H_x = \frac{1}{Z_{e,nm}} E_y \tag{2.37}$$

Each mode will propagate at frequencies above the cutoff frequency:

$$f_{c,nm} = \frac{c}{2\pi} k_{c,nm} = \frac{c}{2}\sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2}$$

and with a propagation constant and wave impedance given by:

$$\gamma_{nm} = j\beta_{nm} = j\sqrt{\left(\frac{2\pi}{\lambda_0}\right)^2 - \left(\frac{n\pi}{a}\right)^2 - \left(\frac{m\pi}{b}\right)^2} \qquad Z_{e,nm} = \frac{\beta_{nm}}{k_0}\eta_0$$

The dominant TE_{10} mode

The lowest possible frequency for which propagation will occur in a rectangular waveguide of dimensions a, b ($a > b$) is obtained from (2.30) putting $n = 1$ and $m = 0$, which corresponds to the TE_{10} mode. Because of this, this mode is called the *dominant mode* (it is the first to appear).

In this case we have:

$$H_z = A\cos\frac{\pi x}{a} e^{-j\beta z} \tag{2.38}$$

and from (2.12) (or from (2.19) and (2.20)) we get the only two remaining non-zero components:

$$H_x = \frac{j\beta}{k_{c,10}} A\sin\frac{\pi x}{a} e^{-j\beta z} \tag{2.39}$$

$$E_y = -\frac{j\beta}{k_{c,10}} Z_h A\sin\frac{\pi x}{a} e^{-j\beta z} \tag{2.40}$$

where $Z_h = \frac{k_0}{\beta}\eta_0 = \frac{\omega\mu_0}{\beta}$ is the wave impedance for the TE_{10} mode (note that we have omitted the subscript 10 for convenience).

The parameters β and k_c are given by:

$$k_c = \frac{\pi}{a} \qquad \beta = \sqrt{k_0^2 - (\pi/a)^2} \tag{2.41}$$

The cutoff frequency is then:

$$f_{c,10} = \frac{c}{2a} \tag{2.42}$$

The next mode to appear will depend on the relative size of a and b.

Note: All previous expressions found for air-filled waveguides (ε_0, μ_0) are also valid for waveguides filled with an homogeneous dielectric of parameters (ε, μ), simply by replacing ε_0 and μ_0 by ε and μ.

Losses in a waveguide

Let's consider a waveguide of arbitrary cross-section but with losses either due to finite wall conductance or to dielectric losses (conductivity in the dielectric). The density of power flow is given by the Poynting vector: $\vec{\mathbf{S}} = \frac{1}{2}\vec{\mathbf{E}} \times \vec{\mathbf{H}}^*$. As the guide has losses, the amplitude of the electric and magnetic field will decay along the direction of propagation z with the factor $\exp(-\alpha z)$. Consequently, the power density will decay with the factor $\exp(-2\alpha z)$ (from the above expression for the Poynting vector).

Now, if P_0 is the total power flow (integral of the power density over the cross-section) at say, $z = 0$, the total transmitted power at a distance $z = z$ is:

$$P(z) = P_0 \exp(-2\alpha z).$$

The rate of decrease of the transmitted power along z must be equal to the total dissipated power per unit length: P_l. That is:

$$P_l = -\frac{\partial P}{\partial z} = 2\alpha P_0 e^{-2\alpha z} = 2\alpha P(z)$$

So the attenuation coefficient α can be expressed as: $\alpha = \frac{P_l}{2P}$

where P, the total power at $z = z$ is given by: $P(z) = \frac{1}{2}\mathrm{Re}\int_S \vec{\mathbf{E}} \times \vec{\mathbf{H}}^* \cdot dS$

Conductor losses in metal walls

These are due to finite conductivity of the waveguide walls. The surface resistance of the walls (per unit width and unit length) is given by $R_m = 1/\sigma\delta\,[\Omega/\mathrm{m}^2]$ where δ is the skin depth.

If the losses are relatively small, a *perturbation method* can be used to calculate them. That is, we assume that the fields in the lossy guide are approximately the same as in a perfect guide, and we can easily calculate these. With this assumption, we can calculate the currents and then use the actual resistance of the walls to calculate the dissipated power. (It is an approximate method since the presence of losses will naturally modify the fields, which we are assuming the same as in the guide with no losses.)

The dissipated power per unit length due to conductor losses is 1/2 the integral of $R_m \vec{\mathbf{J}}_s \cdot \vec{\mathbf{J}}_s^*$ over the complete boundary of the cross-section (as is $RI^2/2$ in a transmission line), that is:

$$P_l = \frac{1}{2} R_m \oint_C \vec{\mathbf{J}}_s \cdot \vec{\mathbf{J}}_s^* dl$$

$\vec{\mathbf{J}}_s$ is calculated as the surface current on the walls of a perfect guide. Then, $\vec{\mathbf{J}}_s = \hat{\mathbf{n}} \times \vec{\mathbf{H}}$, and we have:

$$P_l = \frac{1}{2} R_m \oint_C (\hat{\mathbf{n}} \times \vec{\mathbf{H}}) \cdot (\hat{\mathbf{n}} \times \vec{\mathbf{H}})^* dl = \frac{R_m}{2} \oint_C \vec{\mathbf{H}} \cdot \vec{\mathbf{H}}^* dl$$

The attenuation coefficient due to conductor losses is then given by:

$$\alpha = \frac{R_m \oint_C \vec{\mathbf{H}} \cdot \vec{\mathbf{H}}^* dl}{2\,\mathrm{Re}\int_S \vec{\mathbf{E}} \times \vec{\mathbf{H}}^* \cdot d\vec{S}}$$

This analysis is valid for any waveguide (with arbitrary cross section and mode type) as far as the losses are sufficiently small to be able to assume that the fields will be almost unaffected by the presence of loss.

Example 2.2

Find expressions for the conductor losses and attenuation constant in a coaxial line propagating a TEM mode.

From Example 3.1 we have expressions for the electric and magnetic field as well as for the total power carried by the line:

$$\vec{\mathbf{E}} = \frac{V_0}{\ln(b/a)}\frac{\hat{\mathbf{r}}}{r} \qquad \vec{\mathbf{H}} = \frac{V_0}{\eta\ln(b/a)}\frac{\hat{\phi}}{r} \qquad P = \frac{V_0^2\pi}{\eta\ln(b/a)} \text{ [W]}$$

The power loss due to the finite conductivity of the walls is: $P_l = \frac{R_m}{2}\oint_C \vec{\mathbf{H}}\cdot\vec{\mathbf{H}}^* dl$

where dl is the line element along the perimeter of the cross-section of the conductors.

Then: $P_l = \frac{R_m V_0^2}{2\eta^2(\ln b/a)^2}\left[\oint_{r=a}\frac{r\,d\varphi}{r^2} + \oint_{r=b}\frac{r\,d\varphi}{r^2}\right] = \frac{R_m V_0^2}{2\eta^2(\ln b/a)^2}2\pi\left[\frac{1}{a}+\frac{1}{b}\right]$

and finally, for the attenuation: $\alpha = \frac{R_m\left(\frac{1}{a}+\frac{1}{b}\right)}{2\eta\ln(b/a)}$ [neper/m]

Example 2.3

Power flow and losses in the rectangular waveguide

Considering the dominant TE_{10} mode, the power density in the direction of z is given by:

$$S_z = \frac{1}{2}\vec{\mathbf{E}}\times\vec{\mathbf{H}}^*\cdot\hat{\mathbf{z}} = \frac{1}{2}E_y H_x^*(\hat{\mathbf{y}}\times\hat{\mathbf{x}})\cdot\hat{\mathbf{z}} = -\frac{1}{2}E_y H_x^*$$

and substituting from (2.39) and (2.40):

$$S_z = \frac{1}{2}\vec{\mathbf{E}}\times\vec{\mathbf{H}}^*\cdot\hat{\mathbf{z}} = \frac{\beta}{2k_c^2}k_0\eta_0 A^2\sin^2 k_x x$$

Integrating this over the cross-section we obtain the total power flow along the waveguide:

$$P = \int_0^a\int_0^b \mathrm{Re}\{S_z\}dxdy = \frac{1}{2}\mathrm{Re}\left\{\int_0^a\int_0^b\left(\vec{\mathbf{E}}\times\vec{\mathbf{H}}^*\cdot\hat{\mathbf{z}}\right)dxdy\right\}$$

and evaluating:

$$P = A^2\frac{ab}{4}\frac{\beta k_0}{k_x^2}\eta_0 \tag{2.43}$$

Conduction losses in the guide walls

To consider losses on the guide walls, the currents are approximated by $\hat{\mathbf{n}}\times\vec{\mathbf{H}}$; that is, by the surface current on the walls of a perfect guide ($\hat{\mathbf{n}}$ is the inward-directed normal to the walls). (It is an approximation because the presence of losses will alter the field distribution and this is ignored).

For this mode there are two magnetic field components: H_x and H_z.

On the vertical walls at $x = 0$ and $x = a$ the current is then:

$$\vec{\mathbf{J}}_s = \begin{cases} \hat{\mathbf{x}}\times H_z\hat{\mathbf{z}}\big|_{x=0} = -\hat{\mathbf{y}}A & \text{at } x=0 \\ -\hat{\mathbf{x}}\times H_z\hat{\mathbf{z}}\big|_{x=a} = -\hat{\mathbf{y}}A & \text{at } x=a \end{cases}$$

Now, on the horizontal walls, the surface current is:

$$\vec{\mathbf{J}}_s = \begin{cases} \hat{\mathbf{y}}\times H_x\hat{\mathbf{x}} + \hat{\mathbf{y}}\times H_z\hat{\mathbf{z}} = -\hat{\mathbf{z}}\frac{j\beta}{k_c}A\sin k_x x + \hat{\mathbf{x}}A\cos k_x x & \text{at } y=0 \\ -\hat{\mathbf{y}}\times H_x\hat{\mathbf{x}} - \hat{\mathbf{y}}\times H_z\hat{\mathbf{z}} = \hat{\mathbf{z}}\frac{j\beta}{k_c}A\sin k_x x - \hat{\mathbf{x}}A\cos k_x x & \text{at } y=b \end{cases}$$

The total dissipated power on the walls is: $P_l = \frac{1}{2}R_m\oint_C \vec{\mathbf{J}}_s\cdot\vec{\mathbf{J}}_s^* dl$

per unit length along the z-axis. C is the perimeter of the cross-section of the waveguide and $R_m = 1/\sigma\delta$.

Substituting the currents and integrating we have finally:

$$P_l = R_m A^2\left[b + \frac{a}{2}\left(\frac{\beta}{k_x}\right)^2 + \frac{a}{2}\right] \tag{2.44}$$

The attenuation coefficient due to wall losses is then:

$$\alpha = \frac{P_l}{2P} = \frac{R_m\left[b+\frac{a}{2}\left(\frac{\beta}{k_x}\right)^2+\frac{a}{2}\right]}{\frac{ab}{2}\left(\frac{\beta}{k_x^2}\right)k_0\eta_0} = \frac{R_m\left(2bk_x^2+ak_0^2\right)}{ab\beta k_0\eta_0}\left[\frac{\text{Neper}}{m}\right] \tag{2.45}$$

Attenuation due to dielectric losses

For the propagation constant of a TEM mode propagating in a guide filled with a lossy dielectric, we may simply replace the permittivity by its complex form: $\varepsilon - j\frac{\sigma}{\omega}$, then for TEM modes we have:

$$\gamma_{TEM} = j\omega\sqrt{\mu(\varepsilon - j\sigma/\omega)} \tag{2.46}$$

If the losses are small ($\sigma/\omega\varepsilon \ll 1$), an approximate expression for the attenuation constant (real part of (2.46) is:

$$\alpha_{TEM} \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$$

For TE and TM modes, the propagation constant in a dielectric-filled waveguide has the expression:

$$\gamma_{TE,TM} = \alpha + j\beta = \sqrt{k_c^2 - k^2} \qquad (2.47)$$

If the dielectric has losses, then from the corresponding wave equation (obtained substituting one of Maxwell's curl equations into the other), we have that instead of:

$$k_0^2 = \omega^2 \mu\varepsilon \qquad \text{we have} \qquad k^2 = -j\omega\mu(\sigma + j\omega\varepsilon) = \omega^2\mu\varepsilon\left(1 - j\frac{\sigma}{\omega\varepsilon}\right)$$

using a binomial expansion and replacing in (2.47) gives for the attenuation constant for $\omega > \omega_c$:

$$\alpha_{TE,TM} \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}\left(1 - \frac{\omega_c^2}{\omega^2}\right)^{-1/2} \qquad (2.48)$$

Circular waveguides

Hollow-pipe waveguides of circular cross-sections are also commonly used. They can be attractive for example, when circular polarization is required for the excitation of an antenna. Also, some of the modes of propagation in this waveguide exhibit very low attenuation at high frequencies.

TM modes

In this case, the fundamental equation to solve is:

$$\nabla_t^2 E_z + k_c^2 E_z = 0 \qquad (2.49)$$

and the boundary condition is: $E_z = 0$ at $r = a$

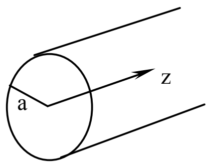

Fig. 2.8 Circular waveguide

The appropriate system of coordinates for this problem is circular cylindrical and (2.49) is then transformed to:

$$\frac{\partial^2 E_z}{\partial r^2} + \frac{1}{r}\frac{\partial E_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 E_z}{\partial \phi^2} + k_c^2 E_z = 0 \qquad (2.50)$$

Using separation of variables again: $E_z = F(r)\, G(\phi)$.

Substituting in (2.50), multiplying by r^2, dividing by FG and re-ordering gives:

$$\frac{r^2}{F}\frac{\partial^2 F}{\partial r^2} + \frac{r}{F}\frac{\partial F}{\partial r} + r^2 k_c^2 = -\frac{1}{G}\frac{\partial^2 G}{\partial \phi^2} \qquad (2.51)$$

The left hand side is only a function of r while the r.h.s. is a function of ϕ. This equation can only be satisfied if both sides are equal to a constant: ν^2.

We have then, two ordinary differential equations:

$$\frac{d^2 G}{d\phi^2} + \nu^2 G = 0 \qquad (2.52)$$

and

$$\frac{d^2 F}{dr^2} + \frac{1}{r}\frac{dF}{dr} + \left(k_c^2 - \frac{\nu^2}{r^2}\right)F = 0 \qquad (2.53)$$

The solution to (2.52) is of the form:

$$G(\phi) = A_1 \cos\nu\phi + A_2 \sin\nu\phi$$

However, the field inside the waveguide must be single-valued; that is: $E_z(\phi) = E_z(\phi + 2\pi)$ and $G(\phi)$ must be periodic with period 2π.

This implies that ν must be an integer: $\nu = n$, $n = 0, 1, \dots$

and the solution for G is then:

$$G(\phi) = A_1 \cos\nu\phi + A_2 \sin\nu\phi\,, \qquad n = 0, 1, \dots$$

Equation (2.53) is called Bessel equation and its solutions are the Bessel functions: $J_n(k_c r)$ and $Y_n(k_c r)$ (of the first and second kind respectively).

Only solutions of the first kind are admissible because $Y_n(k_c r) \to \infty$ as r $\to$ 0.

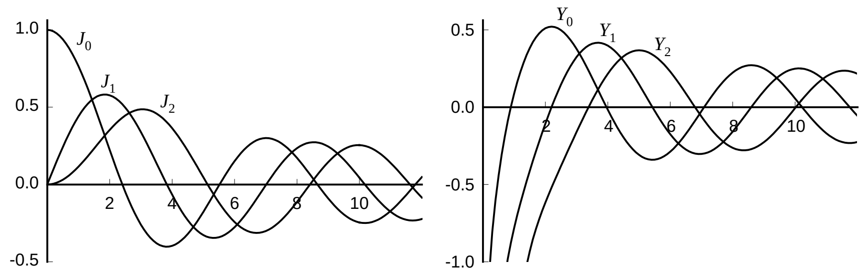

Fig. 2.9 Bessel functions of first and second kind

The general solution for E_z is then:

$$E_z(r,\phi) = \left(A_1 \cos n\phi + A_2 \sin n\phi\right) J_n(k_c r) \qquad (2.54)$$

As the boundary condition is $E_z = 0$ at $r = a$, it is necessary that:

$$J_n(k_c a) = 0 \qquad (2.55)$$

Denoting by p_{nm} the m-th root of the equation $J_n(k_c r) = 0$, the allowed values of k_c are: (in the table the values of p_{nm} for the first three modes are given).

$$k_{c,nm} = \frac{p_{nm}}{a} \qquad (2.56)$$

n	p_{n1}	p_{n2}	p_{n3}
0	2.405	5.520	8.654
1	3.832	7.016	10.174
2	5.135	8.417	11.620

As in the case of the rectangular waveguide, there is an infinite number of possible modes of propagation – each choice of *n* and *m* determines a different mode of propagation: *n* is related to the variation of the field around the circumference and *m*, to the variations in the radial direction.

The propagation constant for the *nm* mode is:

$$\beta = \sqrt{k_0^2 - k_c^2} = \sqrt{k_0^2 - \frac{p_{nm}^2}{a^2}} \tag{2.57}$$

The cutoff frequency and impedance are obtained in similar form as for the rectangular waveguide, giving now:

$$f_c = \frac{1}{\sqrt{\mu\varepsilon}}\frac{p_{nm}}{2\pi a} \quad \text{and} \quad Z_{e,nm} = \frac{\beta_{nm}}{k_0}\eta_0 \quad \text{respectively.}$$

The general solution for TM modes is then:

$$E_z(r,\phi) = J_n(k_c r)\left(A_1 \cos n\phi + A_2 \sin n\phi\right) \tag{2.58a}$$

$$H_r(r,\phi) = -j\frac{nf}{k_c \eta_0 r f_c} J_n(k_c r)\left(A_1 \sin n\phi - A_2 \cos n\phi\right) \tag{2.58b}$$

$$H_\phi(r,\phi) = -j\frac{f}{\eta_0 f_c} J'_n(k_c r)\left(A_1 \cos n\phi + A_2 \sin n\phi\right) \tag{2.58c}$$

$$E_\phi(r,\phi) = -Z_{e,nm} H_r(r,\phi) \tag{2.58d}$$

$$E_r(r,\phi) = Z_{e,nm} H_\phi(r,\phi) \tag{2.58e}$$

($J'_n(k_c r)$ denotes the derivative of $J_n(k_c r)$ with respect to *r*).

TE modes

The basic equation is the same as for TM modes, this time for the H_z component.

The corresponding boundary condition is now: $\dfrac{\partial H_z}{\partial r} = 0$ at $r = a$.

Exercise: Show that this condition is derived from imposing $\hat{r} \cdot \vec{H}_t = 0$ at $r = a$ (the magnetic field is normal to the metal wall).

An appropriate solution for H_z is then:

$$H_z(r,\phi) = J_n(k_c r)\left(B_1 \cos n\phi + B_2 \sin n\phi\right) \tag{2.59}$$

where the condition that determines the value of k_c is: $\dfrac{d J_n(k_c r)}{d r} = 0$ at $r = a$.

Denoting the roots of that equation as p'_{nm}, the allowed values for k_c are:

$$k_{c,nm} = \frac{p'_{nm}}{a} \tag{2.60}$$

n	p'_{n1}	p'_{n2}	p'_{n3}
0	3.832	7.016	10.174
1	1.841	5.331	8.536
2	3.054	6.706	9.970

We can see from (2.56) and (2.60) that the first mode to propagate will be the TE_{11} with a value of k_c equal to 1.841/*a*. The next modes will be in sequence: TM_{01}, TE_{21}, TE_{01} and TM_{11}, TM_{21}, etc.

Planar Transmission Lines

Several forms of waveguiding structures made from parallel metal strips on a dielectric substrate are used extensively in microwave and millimetre-wave circuits. The most common types are the *stripline*, the *microstrip* and the *coplanar waveguide*. These guides are operated in the lowest mode which is a TEM in the case of the stripline and a quasi-TEM for the microstrip and the coplanar guide.

Stripline

The stripline consists of a conductive strip lying between two parallel, wide conductive planes, as shown in the figure. The region between the plates is filled with a uniform dielectric. Such a structure, with a uniform dielectric and more than one conductor can support a TEM wave. If the strip width *w* is much grater than the spacing *d*, the structure can be approximated by two parallel-plate waveguide connected in parallel, but a better approximation is obtained using the capacitance per unit length of the structure. For a TEM wave, the phase velocity is $v_p = (\mu\varepsilon)^{-1/2}$ and using the transmission line formalism, $v_p = (LC)^{-1/2}$, In the same form, for the characteristic impedance: $Z_0 = \sqrt{\dfrac{L}{C}} = \dfrac{\sqrt{LC}}{C} = \dfrac{\sqrt{\mu\varepsilon}}{C}$ and the characteristic impedance can be found from the capacitance.

There are no analytic expressions possible in this case to find these values and a series of empirical relations have been developed to help in the design of these structures. Better solutions are found using numerical techniques.

Microstrip

The most widely used of these guides is the microstrip which consists of one conductive strip lying on top of a dielectric slab backed with a conductive (ground) plane. The dielectric above the strip (usually air) is different to the slab, so the structure cannot support truly TEM waves. However, the lowest order of propagation is a mode very similar to a TEM mode, a *quasi-TEM* wave and a quasistatic approach is normally used to derive expressions for this waveguide. This approximation is based on the estimation of an *effective permittivity*, which is the permittivity of a uniform medium surrounding the strip.